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INDIVIDUAL ASSIGNMENT

LEVEL 05

COMP50059

Artificial Intelligence

AI2561

Hand Out Date: As Specified in SAIS

Hand In Date: As Specified in SAIS

INSTRUCTION TO CANDIDATES

1. **Students are advised to underpin their answers with the use of references (cited using the Harvard Referencing Style).**

2. **Late submission will be awarded zero (0) unless extenuating circumstances (EC) are upheld.**

3. **Cases of plagiarism will be penalized.**

4. **The assignment should be submitted as a softcopy:**

   a. **The softcopy of the written assignment and source code must be uploaded to given link in the LMS.**

Module Code	**COMP50059**
Title of Assignment	**Assignment 2 -** Hotel Management Forecasting and Decision Support
Assignment Weighting	**70%**
Submission Date	13th Week

Learning outcomes

1. Demonstrate a critical understanding of Artificial Intelligence (AI) applications, Machine Learning (ML) techniques and Data Mining (DM) approaches through real world examples (Knowledge and Understanding, Learning, Enquiry)

2. Analyse a given task scenario, define the business problem and select an appropriate ML technique and DM approach to implement an AI solution to a given system task and critically appraise the solution (Analysis, Reflection, Problem Solving)

3. Implement a ML model, and presenting logical and coherent written arguments for choices made in relation to ML technique and DM approaches (Problem Solving, Reflection, Application)

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1. Fundamental requirements

In this assignment, you must satisfy all the following fundamental requirements, otherwise you could be awarded zero marks for the assignment.

- You are expected to explain the concepts in your own words or rephrase and avoid quotes. You are referred to the University's regulations on plagiarism available at https://www.staffs.ac.uk/students/course-administration/academic-policies-and-regulations/academic-conduct-procedure If the Turnitin system detects plagiarism, it will be reported to the school and appropriate actions will be taken.
- You must be the only author of the work you submit for assessment. You must not have help with this assignment from any person except for members of the regular teaching team. You are reminded of the university's policy about academic misconduct, as described at: http://www.staffs.ac.uk/support_depts/info_centre/handbook/academic-life.jsp
- You must submit all parts of your assignment via this module's Blackboard presence using the links provided in the Assessment section. Follow the instructions given with each link.
- During the demonstration, you must be able to explain in detail the software code you present for assessment.

2. Submission Requirements

a. You are required to submit a Python Jupyter Notebook (.ipynb) as your main deliverable, containing all your code, results, explanations, and discussions in well- organised code and markdown cells. The notebook must demonstrate the full workflow, including data preprocessing, model building, evaluation, comparisons, and reflections. In addition, you have the option to submit a short supplementary report (Word, maximum 1500 words) if you feel certain aspects such as model comparison, fine-tuning, or business implications, require further explanation beyond what is practical in the notebook. However, the majority of your explanations should remain in the notebook itself. To improve the document readability and reduce the number of words, you can use graphs, figures etc. Word count guidance is available at: https://askus.staffs.ac.uk/academicskills/faq/252665

b. You must ensure **academic integrity** in all submitted work, including responsible and transparent use of AI tools, and appropriately acknowledge any AI assistance. For guidance on academic integrity and use of AI, please refer to the library resource: Home - AI - Library at University of Staffordshire.

c. It is strongly recommended that you attempt to submit a bit early to avoid any technical difficulty or missing the deadline.

d. If you are unsure about academic writing, references etc, the library services team offer excellent 1 to 1 support and group support https://libguides.staffs.ac.uk/study-skills This is a free service you just need to make an appointment.

3. Assignment Details

This is a practical, hands-on assessment where you will complete five tasks, each focusing on the development and application of different machine learning techniques and algorithms. You will be provided with a scenario to guide your work and context for completing the tasks. You will use Python as the programming language and Google Colab as your development environment. Your work will be evaluated based on the submitted artefacts for each task as well as a demonstration of your solutions.

3.1 Here are some key points to keep in mind:

- Tasks: There will be multiple tasks for you to complete, each designed to assess your skills in data mining tasks to develop different machine learning techniques and algorithms.
- Task Order: You are not required to complete the tasks in a specific order. You can work on them in any sequence you prefer. You can switch between tasks as needed to complete them. Take advantage of this flexibility to manage your time effectively and address tasks based on your strengths and preferences.
- Completion: Ensure that you thoroughly understand each task and complete it to the best of your ability. Pay attention to the specific requirements and guidelines provided for each task.

3.2 Scenario

An international online retail company, ShopSphere, has experienced rapid growth in customer acquisition over the past few years. However, despite increasing user registrations, the company is facing challenges in customer retention, revenue consistency, and personalized marketing effectiveness.

The management team has identified several critical business concerns:

- A growing number of customers are becoming inactive or leaving the platform (customer churn)
- Difficulty in accurately predicting customer spending and lifetime value
- Lack of clear understanding of different customer segments and their behavior patterns

To address these challenges, ShopSphere has collected historical customer data, including demographic details, transaction behavior, and engagement metrics.

You have been hired as a Junior Data Scientist to analyse e-commerce behavioral data and develop machine learning solutions to support business decision-making.

The company wants to:

- Reduce Customer Churn
- Predict Customer Spending
- Segment customers

Your task is to apply classification, regression, and clustering techniques to generate actionable insights. You will also demonstrate data preprocessing, feature engineering, model evaluation, and visualization skills.

You Must:

1. Follow the CRISP-DM methodology (Business Understanding, Data Understanding Data Preparation, Modelling, Evaluation, Deployment/Insights)
2. Preprocess the dataset correctly: handle missing values, encode categorical variables, normalise numeric features, and create meaningful new features
3. Apply at least three machine learning algorithms for classification tasks
4. Apply at least three machine learning algorithms for regression tasks
5. Apply at least three clustering algorithm and interpret the results
6. Visualise patterns and model performance clearly
7. Translate findings into business recommendations.

3.3 Dataset Overview

- Data source
  For this assignment, you are required to use the E-Commerce Customer Behavior, which can be accessed here:

  https://www.kaggle.com/datasets/dhairyajeetsingh/ecommerce-customer-behavior-dataset

- Features

1. Customer Demographics (5 features)

- Age, Gender, Country, City, Membership_Years

2. Platform Engagement (8 features)

- Login_Frequency, Session_Duration_Avg, Pages_Per_Session
- Cart_Abandonment_Rate, Wishlist_Items, Email_Open_Rate
- Mobile_App_Usage, Social_Media_Engagement_Score

3. Purchase Behavior (6 features)

- Total_Purchases, Average_Order_Value, Days_Since_Last_Purchase
- Discount_Usage_Rate, Return_Rate, Payment_Method_Diversity

4. Customer Service (3 features)

- Customer_Service_Calls, Product_Reviews_Written, Lifetime_Value

5. Financial & Status (3 features)

- Credit_Balance, Churned (target variable), Signup_Quarter

3.4 Assignment Tasks

Task 1: Data Preprocessing

The data mining phase involves preparing your data before training and testing your ML models. For Tasks 2, 3, and 4, create a dedicated section in each notebook to implement task-specific preprocessing steps. For example, Task 2 (Classification) may require normalization and label encoding, while Tasks 3 and 4 may need different processing steps.

Data Understanding: to understand the characteristics of the dataset, including data types, distributions, missing values, and outliers.

Data Preparation:

- Pre-process the dataset by handling missing values, outliers, and feature scaling as necessary.

a. Example: for missing values,
   - ✓ Check if numeric (children, babies): replace missing with 0
   - ✓ Check if categorical (country, agent, company): Fill with "unknown" or drop rows (justify your choice)

b. Example: for normalisation / scale Numeric features:
   - ✓ Numeric columns:
   - ✓ Apply scaling: StandardScaler (mean=0, std=1) or MinMaxScaler ([0,1])

Split the dataset into training and testing sets for model evaluation.

Encode Categorical Variables: Identify categorical features:

a. Encode for ML:
   - ✓ **One-Hot Encoding**: Non-ordinal categories (e.g. hotel type, meal, market segmentation)
   - ✓ **Label Encoding**: Ordinal categories (e.g. month number if using numeric)

Feature Engineering

a. Engagement Score**:**

```
engagement_score = login_frequency + session_duration_avg +
pages_per_session + mobile_app_usage + social_media_engagement_score
```

b. Purchase Activity Index**:**

```
purchase_activity = total_purchases / (membership_years + 1)
```

(+1 added to avoid division by zero and stabilize new users)

c. Customer Risk Score**:**

```
risk_score = cart_abandonment_rate + return_rate +
customer_service_calls + discount_usage_rate
```

Optional Derived Features

a. Total Engagement Score:

```
total_engagement = login_frequency + session_duration_avg +
pages_per_session + mobile_app_usage + social_media_engagement_score
```

(captures overall customer interaction intensity across platforms)

b. Purchase Frequency Rate:

```
purchase_frequency = total_purchases / (membership_years + 1)
```

(normalizes buying behavior based on customer lifespan)

c. Customer Friction Index:

```
friction_index = cart_abandonment_rate + return_rate +
customer_service_calls
```

(measures dissatisfaction and potential churn pressure signals)

Task 2: Classification Task – Churn prediction

a. Target: churned
b. Input features: Age, Gender, Membership_Years, Login_Frequency, Session_Duration_Avg, Cart_Abandonment_Rate, Mobile_App_Usage, Total_Purchases, Average_Order_Value, Days_Since_Last_Purchase, Return_Rate, Customer_Service_Calls, Lifetime_Value
c. Train the classification models using the training data.
d. Evaluate its performance on the validation and testing data using relevant evaluation metrics.
e. Fine-tune the classification model parameters to optimize performance.
f. Utilize evidence and examples from your experimental results to support your comparison of algorithms performance and conclusions.

Note: Business insight: Identify high-risk customers using churn predictions to enable targeted retention strategies, protect high-value customers, improve engagement, and enhance service quality to reduce customer loss and increase lifetime value.

Task 3: Regression Task – Lifetime value forecasting

a. Target: Lifetime_Value
b. Input features: Age, Gender, Membership_Years, Login_Frequency, Session_Duration_Avg, Cart_Abandonment_Rate, Mobile_App_Usage, Social_Media_Engagement_Score, Total_Purchases, Average_Order_Value, Days_Since_Last_Purchase, Discount_Usage_Rate, Return_Rate, Customer_Service_Calls, Credit_Balance
c. Evaluation metrics: RMSE, MAE, R^2

Note: Business insight: Forecast customer lifetime value to optimize revenue strategies, improve targeted marketing, and identify high-value customers for retention and personalized pricing decisions.

Task 4: Clustering Task – Segment Customers / Bookings

a. Target: None (unsupervised)
b. Input features: Age, Gender, Country, Membership_Years, Login_Frequency, Session_Duration_Avg, Pages_Per_Session, Cart_Abandonment_Rate, Mobile_App_Usage, Social_Media_Engagement_Score, Total_Purchases, Average_Order_Value, Discount_Usage_Rate, Return_Rate, Days_Since_Last_Purchase, Customer_Service_Calls, Credit_Balance, Lifetime_Value
c. Evaluation metrics: Silhouette Score, Davies-Bouldin Index

Business insight: Identify meaningful customer segments based on behavior, engagement, and value patterns to enable targeted marketing campaigns, personalized offers, and improved customer retention strategies.

Task 5: Evaluation & Reporting

For Tasks 2, 3, and 4, add a new section titled "Evaluation and Reporting". This section should include the following:

a. Compare models and explain hyperparameters
b. Visualize patterns (heatmaps, cluster plots, cancellation rates by segment)
c. Translate results into business recommendations (staffing, pricing, marketing)

4. Instructions for the demonstration

You can find your demonstration appointment time will be booked in within the last week of teaching. To ensure a smooth process, it is crucial that you arrive at least ten minutes early for your appointment and complete the following steps before your scheduled start time:

1. Log on to the pc.
2. Open and prepare your code.
3. Verify that your processes are ready to run without delay.
4. Await instructions from the lecturer who will assess your demonstration.

Your technical understanding and skills will be assessed by you doing a walkthrough of your code and then be asked a set of questions. For the demonstration, you will only need your code. You will not need to prepare any other documentation.

Failure to demonstrate your software will result in zero marks for this assignment.

5. Submit your work

On or before the submission deadline date, upload following files:

- Your report (word file)
- Your AI artefact: one **ZIP file** containing all processes for your ML models.
  1. Give an appropriate name to your file. For instance, for Task 2, create a file named "Classification".
  2. Repeat the same steps for Task 3 and Task 4
  3. Save all tasks inside a main folder named “Assignment 2”.
  4. Compress the “Assignment 2” folder into a zip file.
  5. Upload the zip file

Note: Your code files should be saved as **Jupyter Notebook (.ipynb) files**.

6. Marking Criteria

You should refer to the rubric during your work on this assignment. The rubric shown below indicates how the marks will be allocated

Criterion (Weighting)	Fail	Third	Lower Second 2:2	Upper Second 2:1	First Class		
	(0–39)	(40–49)	(50–59)	(60–69)	(70–79)	(80–89)	(90–100)
Data Understanding & Preparation (15%)	No meaningful exploration. No handling of missing values/encoding.	Minimal preprocessing with major errors.	Basic preprocessing: some missing values handled, limited encoding/scaling. Weak feature engineering.	Mostly correct preprocessing (missing values, encoding, scaling). Some feature engineering (guests/nights).	Full preprocessing (missing values, encoding, scaling). Good feature engineering including seasonal features.	Excellent preprocessing with strong justification. Advanced feature engineering with clear rationale.	Outstanding preprocessing with professional-level EDA. Feature engineering innovative and highly relevant to hotel operations.
Coding Quality & Notebook Structure (10%)	Notebook incomplete/unreadable. Poor or no code commenting.	Minimal coding effort. No structure or explanation.	Basic coding structure. Few comments, code not reusable.	Code structured reasonably. Some comments, functions partially used.	Good coding practices. Clear comments, structured notebook with logical flow.	Excellent coding quality. Functions, reusable components, modular approach. Notebook clear and well-explained.	Professional-level notebook. Highly readable, polished, reusable code with excellent narrative and documentati

							on.
Model Building and Evaluation **(60%)**	Machine learning models not trained or evaluated.	A few machine learning models are trained, evaluated, and compared with limited proficiency. Performance assessment lacks depth.	Models for Tasks 1, 2, and 3 are included. All models can be executed, and the results are clearly displayed. The learner can demonstrate a walkthrough of the following stages based on one of the tasks: • Data preparation • Model building • Model evaluation and optimisation	Models for Tasks 1, 2, and 3 are included. All models can be executed, and the results are clearly displayed. The learner demonstrates excellent knowledge by meeting the following criteria: • Evaluation of challenges faced during the data preparation stage and how they were overcome. • Evaluation of parameters used in the model building process.	Models for Tasks 1, 2, and 3 are included. All machine learning models are trained, evaluated, and compared with exceptional proficiency. The learner's walkthrough shows outstanding knowledge of the following stages, based on the tasks. This includes all conditions listed in the previous column, plus the additional criteria below: • Models are not only compared but critically discussed, with pros/cons. • Overfitting/underfitting diagnosed and addressed (via tuning, regularization, early stopping). • Use of visualisations to support evaluation (ROC curves, residual plots, cluster heatmaps). •		

				• Evaluation of model evaluation metrics and optimisation techniques. • Cross-validation used for robust evaluation. • Regression: Advanced techniques attempted: regularization (Lasso/Ridge/ElasticNet) or ensemble regression. Evaluated with RMSE, MAE, R^2, and results compared. • Clustering: Optimal cluster number determined	

				systematically (Elbow method, silhouette analysis). • Classification: Models evaluated with multiple metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC).	
Response to Questions **(5%)**	No answer to questions	Provides some accurate and relevant answers			Provides accurate and relevant answers. Answers show depth of knowledge and insight. Engages confidently with the assessor
Academic Integrity and AI **(10%)**	Heavy reliance on AI-generated content with no adaptation or understanding. Work is largely copy-paste. No evidence of student's own critical thinking.	AI used selectively and responsibly. Student adapts outputs substantially, integrates them into their own workflow, and explains decisions. Original contribution is clear.			Outstanding originality with minimal reliance on AI. Student demonstrates deep understanding, goes beyond AI suggestions, and produces a highly polished and creative piece of work. AI may be used for minor support only (e.g., code debugging), with full acknowledgment and critical reflection.